

Dynamically decoupled radio-frequency(DDrf) Gate

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Outline

1. DDrf Gate

- Hamiltonian engineering & conditional rotation
- DDrf spectroscopy

2. Side-Peak Problem

- Observation in numerics
- Detuned-Rabi explanation

3. Suppression

- Apodized pulses & window comparison
- Outlook: time-dependent Ω_{RF} via Magnus expansion

DDrf: Hamiltonian Engineering

DDrf = selective, phase-controlled RF driving of nuclear spins, interleaved with dynamical decoupling on the electron spin.

If the Hamiltonian is **block-diagonal** in the electron basis,

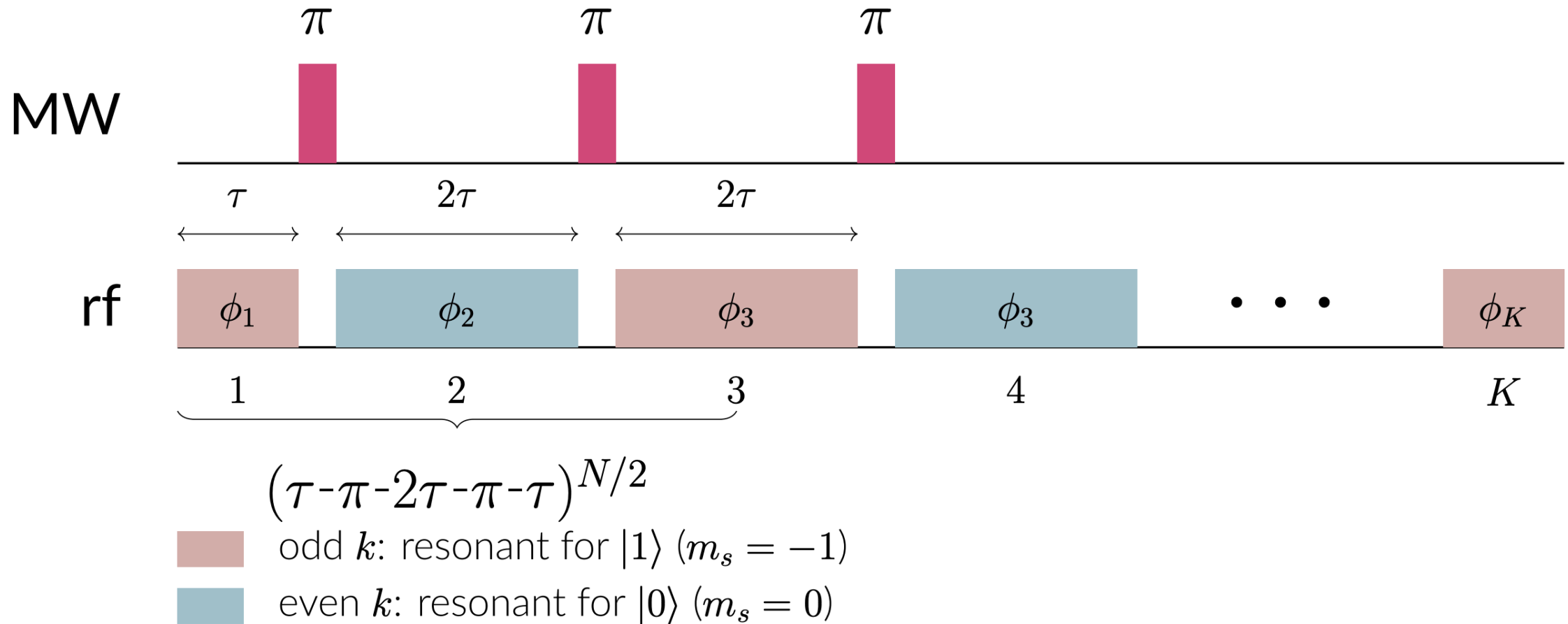
$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1,$$

then sandwiching evolution with an electron π -pulse swaps the two branches:

$$|+\rangle|0\rangle \xrightarrow{t_1, \pi, t_2} \frac{1}{\sqrt{2}} \left(|0\rangle \otimes \underbrace{e^{-iH_0t_2} e^{-iH_1t_1}}_{U_0} |0\rangle + |1\rangle \otimes \underbrace{e^{-iH_1t_2} e^{-iH_0t_1}}_{U_1} |0\rangle \right).$$

Generally $U_0 \neq U_1$ — a **conditional gate**.

DDrf: Pulse Sequence



DDrf Spectroscopy: NV- ^{13}C Hamiltonian

For NV- ^{13}C with rf driving:

$$H = |0\rangle\langle 0| \otimes H_0 + |-1\rangle\langle -1| \otimes H_1$$

$$H_0 = \omega_0 I_z + 2\Omega_{\text{RF}} \cos(\omega_{\text{RF}}t + \phi) I_x$$

$$H_1 = \omega_1 \tilde{I}_z + 2\Omega_{\text{RF}} \cos \beta \cos(\omega_{\text{RF}}t + \phi) \tilde{I}_x + 2\Omega_{\text{RF}} \sin \beta \cos(\omega_{\text{RF}}t + \phi) \tilde{I}_z$$

The $|1\rangle$ branch has a **tilted** nuclear quantization axis (angle β , with $\sin \beta = A_{\perp}/\omega_1$). This small tilt is what enables hyperfine-mediated control of nuclei with vanishing $A_{\perp}$.

DDrf Spectroscopy: Rotating Frame

Two electron-conditioned rotating frames, $R_s(t) = e^{i\omega_{\text{RF}}t} \tilde{I}_z^{(s)}$, give:

$$H'_0 = (\omega_0 - \omega_{\text{RF}}) I_z + \Omega_{\text{RF}} (\cos \phi I_x + \sin \phi I_y)$$

$$H'_1 = (\omega_1 - \omega_{\text{RF}}) \tilde{I}_z + \Omega_{\text{RF}} \cos \beta (\cos \phi \tilde{I}_x + \sin \phi \tilde{I}_y)$$

At resonance ($\omega_{\text{RF}} = \omega_1$) and approximation to $\omega_0 - \omega_{\text{RF}} \gg \Omega_{\text{RF}}$ and $\beta \rightarrow 0$, H'_1 is a pure transverse drive, while H'_0 is pure Z-rotation.

DDrf Spectroscopy: Full Time Evolution

Each MW π -pulse acts as an **instantaneous frame swap** $\Lambda_{s,\bar{s}}(t) \equiv R_s(t)R_{\bar{s}}(t)^\dagger$ between the two rotating frames. The full unitary $U = \sum_s |s\rangle\langle s| \otimes U_s$ is then a chain of H'_s -segments separated by frame swaps:

$$U_s = R_s(4N\tau)^\dagger \cdot e^{-iH'_s\tau} \cdot \Lambda_{s,\bar{s}}((2N-1)\tau) \cdot e^{-iH'_{\bar{s}}2\tau} \cdot \Lambda_{\bar{s},s}((2N-3)\tau) \cdot e^{-iH'_s\tau} \dots \\ \dots e^{-iH'_s\tau} \cdot \Lambda_{s,\bar{s}}(3\tau) \cdot e^{-iH'_{\bar{s}}2\tau} \cdot \Lambda_{\bar{s},s}(\tau) \cdot e^{-iH'_s\tau} \cdot R_s(0).$$

Exact under the assumption of negligible MW pulse duration, and **much faster than direct Schrödinger integration** when Ω_{RF} is constant — each $e^{-iH'_s\tau}$ is a single matrix exponential of a time-independent Hamiltonian.

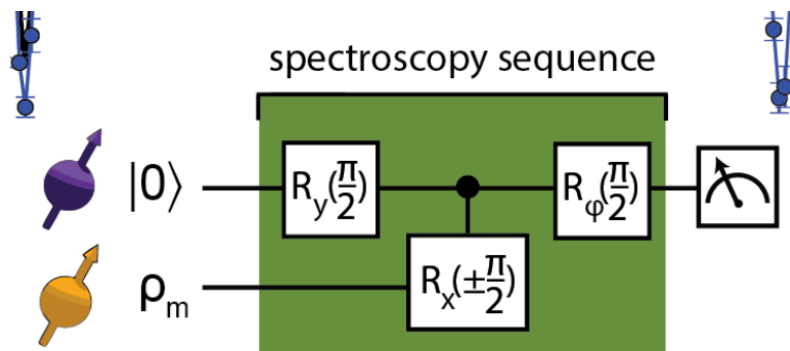
DDrf Spectroscopy: Procedure & Results

Sequence: $\pi/2 \rightarrow \text{DDrf}(N, \tau) \rightarrow \pi/2_\phi$, projecting onto $|+\rangle$ ($\phi = \pi/2$).

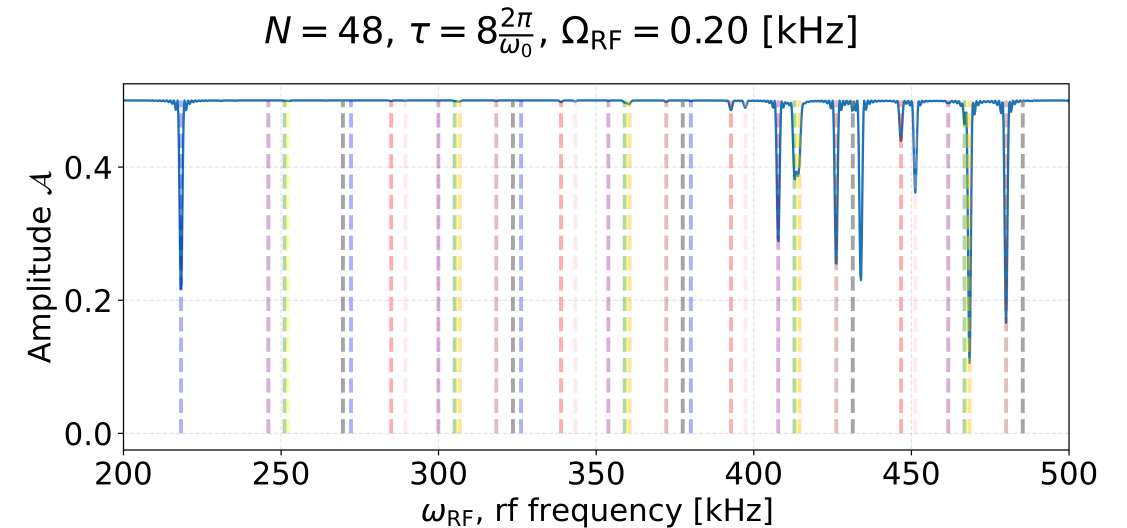
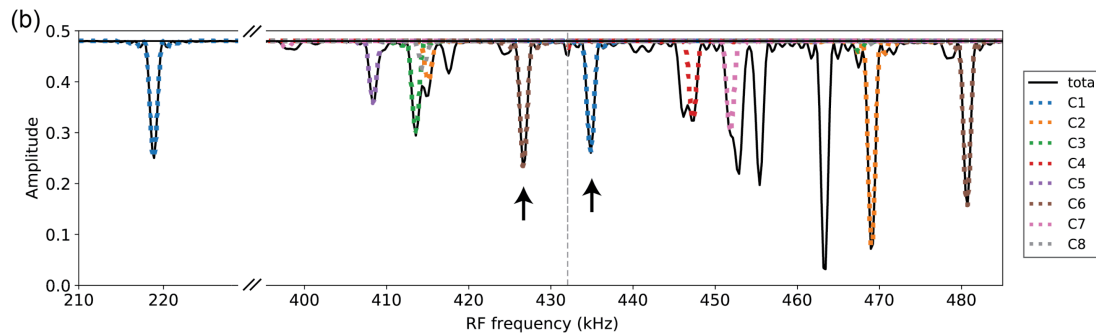
For N nuclear spins:

$$P_x = \frac{1}{2} + \frac{1}{2^{N+1}} \Re \text{Tr } U_0 U_1^\dagger, \quad \text{Tr } U_0 U_1^\dagger = \prod_i \text{Tr } U_0^i U_1^{i\dagger}.$$

Peaks appear at $\omega_{\text{RF}} = \omega_1$.



DDrf Spectroscopy: Reproduced Taminiau Results

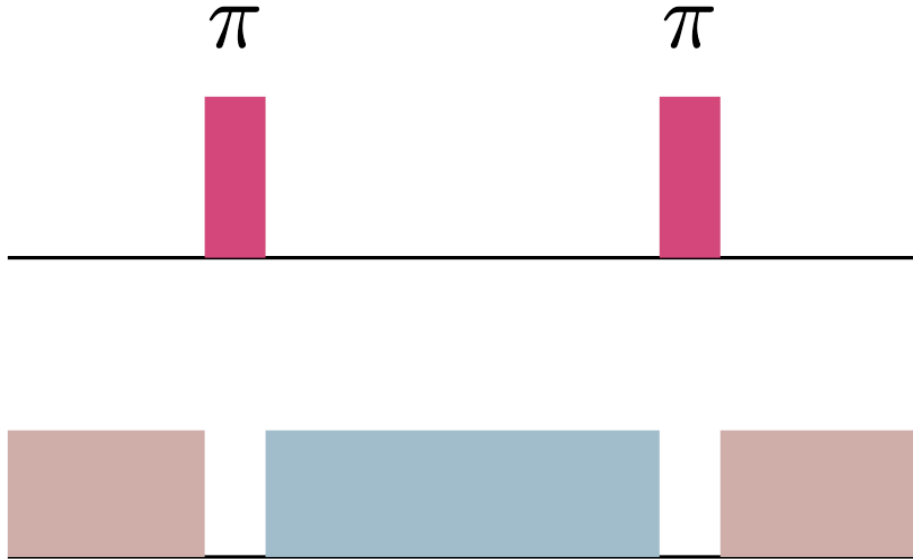


Left: Taminiau et al., *Phys. Rev. X* **9**, 031045 (2019). Right: our numerics — peak positions and amplitudes match.

DDrf: Per-Cell Decomposition

The DDrf sequence is built from identical 4- τ cells. Each cell is itself block-diagonal:

$$V^{(k)} = |0\rangle\langle 0| \otimes V_0^{(k)} + |1\rangle\langle 1| \otimes V_1^{(k)}, \quad U = \sum_{s \in \{0,1\}} |s\rangle\langle s| \otimes \prod_{k=1}^{N/2} V_s^{(k)}.$$



DDrf: Telescoping Strategy

A z -rotation can be commuted *through* a transverse rotation by shifting its azimuthal angle:

$$e^{i\alpha I_z} (\cos \phi I_x + \sin \phi I_y) e^{-i\alpha I_z} = \cos(\phi - \alpha) I_x + \sin(\phi - \alpha) I_y.$$

Applied per cell (Taminiau limit, $\beta = 0$, $\omega_{\text{RF}} = \omega_1$):

$$V_0^{(k)} = e^{-iH'_0\tau} e^{-iH'_1 2\tau} e^{-iH'_0\tau} = e^{-i 2\delta_0\tau I_z} e^{-i 2\Omega\tau \hat{\phi}'_k \cdot \vec{I}}.$$

Each cell splits cleanly into a z -piece + a transverse rotation. Choosing ϕ_k to align successive transverse axes makes the product over k **telescope** into a single conditional rotation.

Hybrid DDrf: Two Limits and Their Combination

The per-cell picture exposes two distinct mechanisms that both produce conditional rotation:

- **CPMG limit** ($\Omega_{\text{RF}} \rightarrow 0$): pure dynamical decoupling. At $\tau \simeq \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$, the change-of-frame between R_0 and R_1 alone yields a hyperfine-driven conditional rotation.
- **Taminiau limit** ($\beta \rightarrow 0$, $\omega_{\text{RF}} = \omega_1$): pure RF driving. Conditional rotation comes from Ω_{RF} alone, independent of $A_{\parallel}$.

Hybrid idea. Choose τ at the CPMG resonance and add RF driving on top. The two mechanisms add coherently:

$$\theta_{\text{cond}} \approx \underbrace{N \theta_{\text{CPMG}}(\tau)}_{\text{frame-swap}} + \underbrace{N \Omega_{\text{RF}} \tau}_{\text{RF drive}}$$

Hybrid DDrf: Taylor Expansion in β

The hybrid regime sits *between* the two clean limits, but β is small. Treat it as a perturbation around the Taminiau limit:

$$V_s^{(k)} \simeq V_s^{(k)} \Big|_{\beta=0} + \beta \frac{\partial}{\partial \beta} V_s^{(k)} \Big|_{\beta=0} + \mathcal{O}(\beta^2).$$

Limit checks (the reason this approach is trustworthy):

- $\beta \rightarrow 0$ — recovers the Taminiau result (slide 10).
- $\Omega \rightarrow 0$ — recovers the CPMG mechanism.

Outcome (qualitative). The leading β -correction adds a small k -dependent rotation around an axis that mixes I_z, I_x, I_y . With an appropriate choice of ϕ_k and τ , these corrections **telescope coherently** — the effective rotation angle exceeds bare $\Omega\tau$, realizing the speed-up promised on the previous slide.

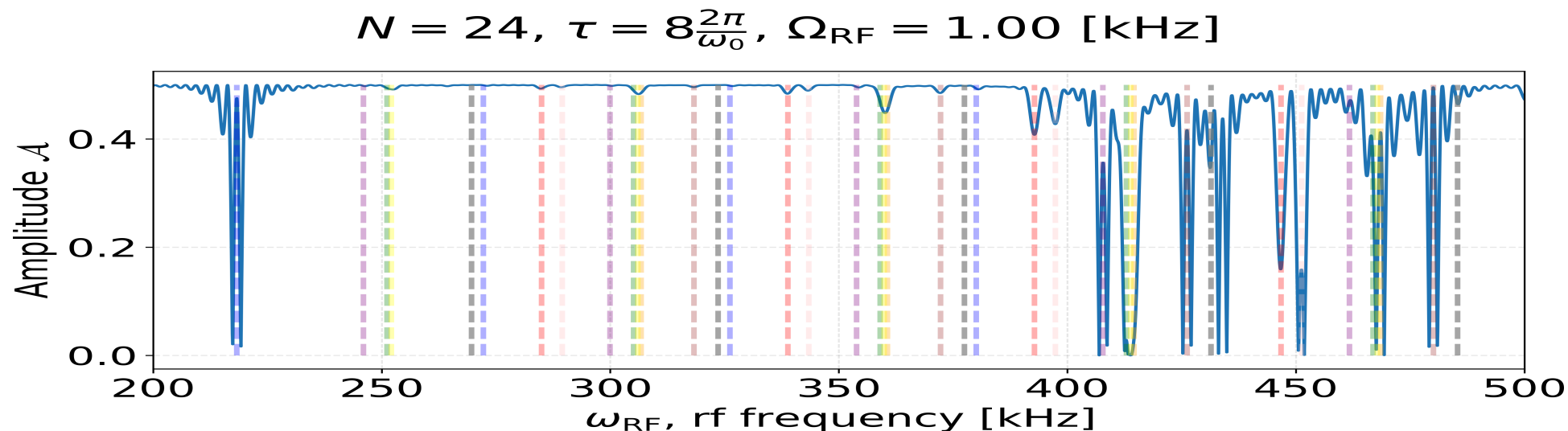
Side-Peak Problem: Observation

In the Taminiau limit ($\beta \rightarrow 0, \omega_{\text{RF}} = \omega_1$), telescoping gives

$$U_s = R_z(N(\omega_L - \omega_1)\tau) \cdot R_\phi(\pm N\Omega_{\text{RF}}\tau).$$

[Observation] When $N\Omega_{\text{RF}}\tau = 2\pi$, $U_0 = U_1$ — the gate becomes **unconditional**, so a flat spectroscopy signal is expected.

The numerics disagree:



Side-Peak Problem: Detuned Rotating Frame

Restore finite detuning $\delta_1 = \omega_1 - \omega_{\text{RF}}$ (assume $\beta = 0$ for clarity):

$$H'_1 = \delta_1 I_z + \Omega(\cos \phi I_x + \sin \phi I_y) = \Omega_{\text{eff}} \hat{n}(\phi) \cdot \vec{I},$$

$$\Omega_{\text{eff}} = \sqrt{\Omega^2 + \delta_1^2}, \quad \sin \gamma = \frac{\delta_1}{\Omega_{\text{eff}}}, \quad \hat{n}(\phi) = (\cos \gamma \cos \phi, \cos \gamma \sin \phi, \sin \gamma).$$

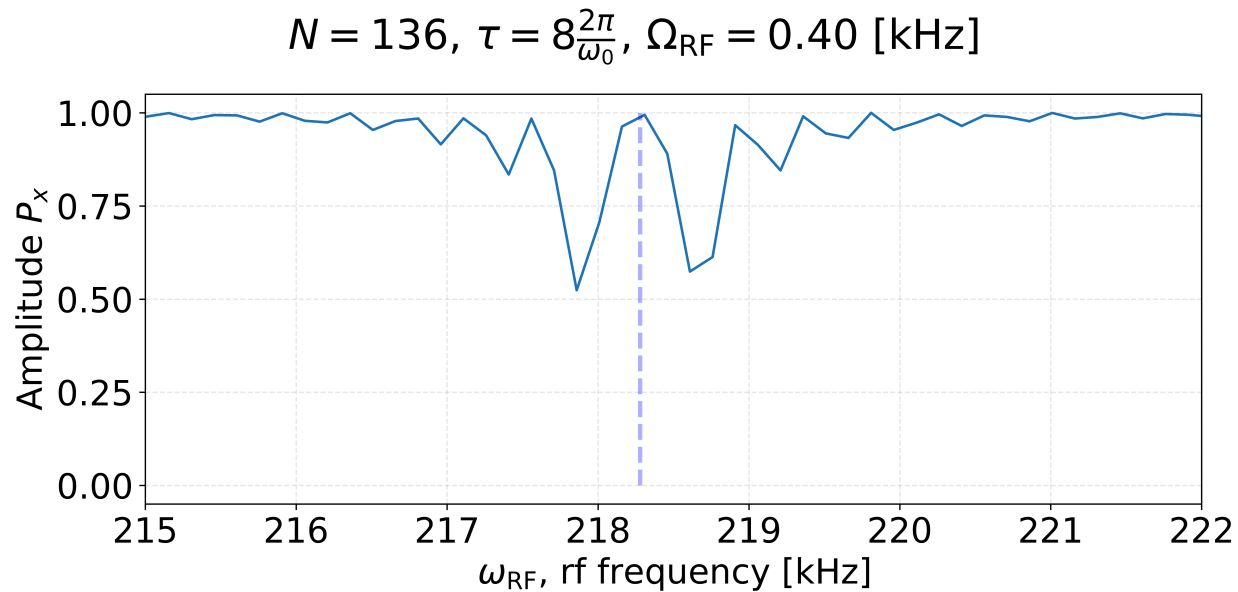
Conjugation/telescoping still works because the tilt angle γ is invariant under z -rotations.
Result:

$$V_s^{\text{tot}} = e^{-iN\delta_0\tau I_z} e^{-i\Omega_{\text{eff}}N\tau \hat{n}_s \cdot \vec{I}}, \quad \hat{n}_{0,1} = (\pm \cos \gamma, 0, \sin \gamma).$$

Side-Peak Problem: Detuned Rabi Formula vs. Numerics

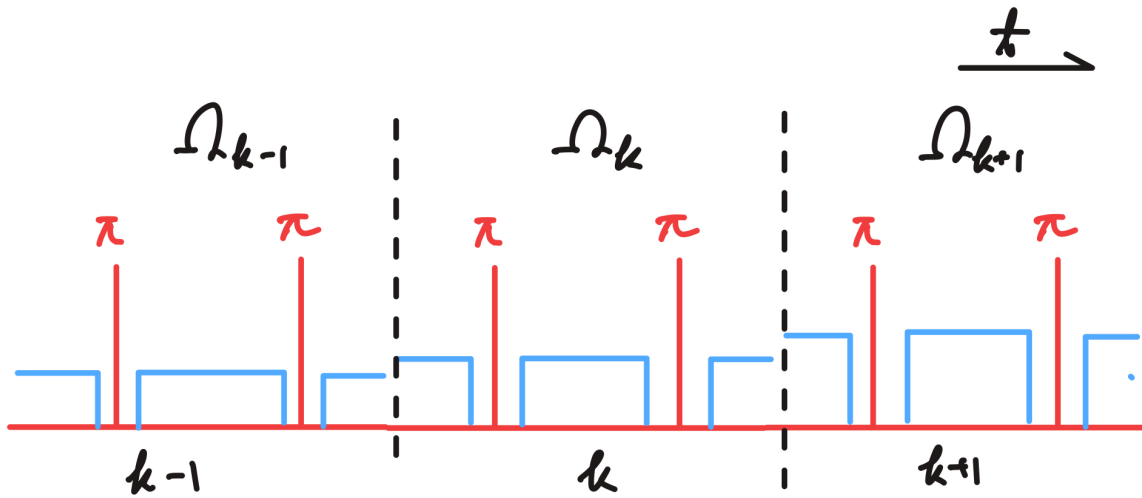
$$\frac{1}{2} \text{Tr}(U_0 U_1^\dagger) = 1 - \frac{2\Omega^2}{\Omega^2 + \delta_1^2} \sin^2 \left(\frac{\sqrt{\Omega^2 + \delta_1^2} N\tau}{2} \right)$$

The detuned-Rabi formula reproduces both the envelope and the side-lobe period observed numerically:



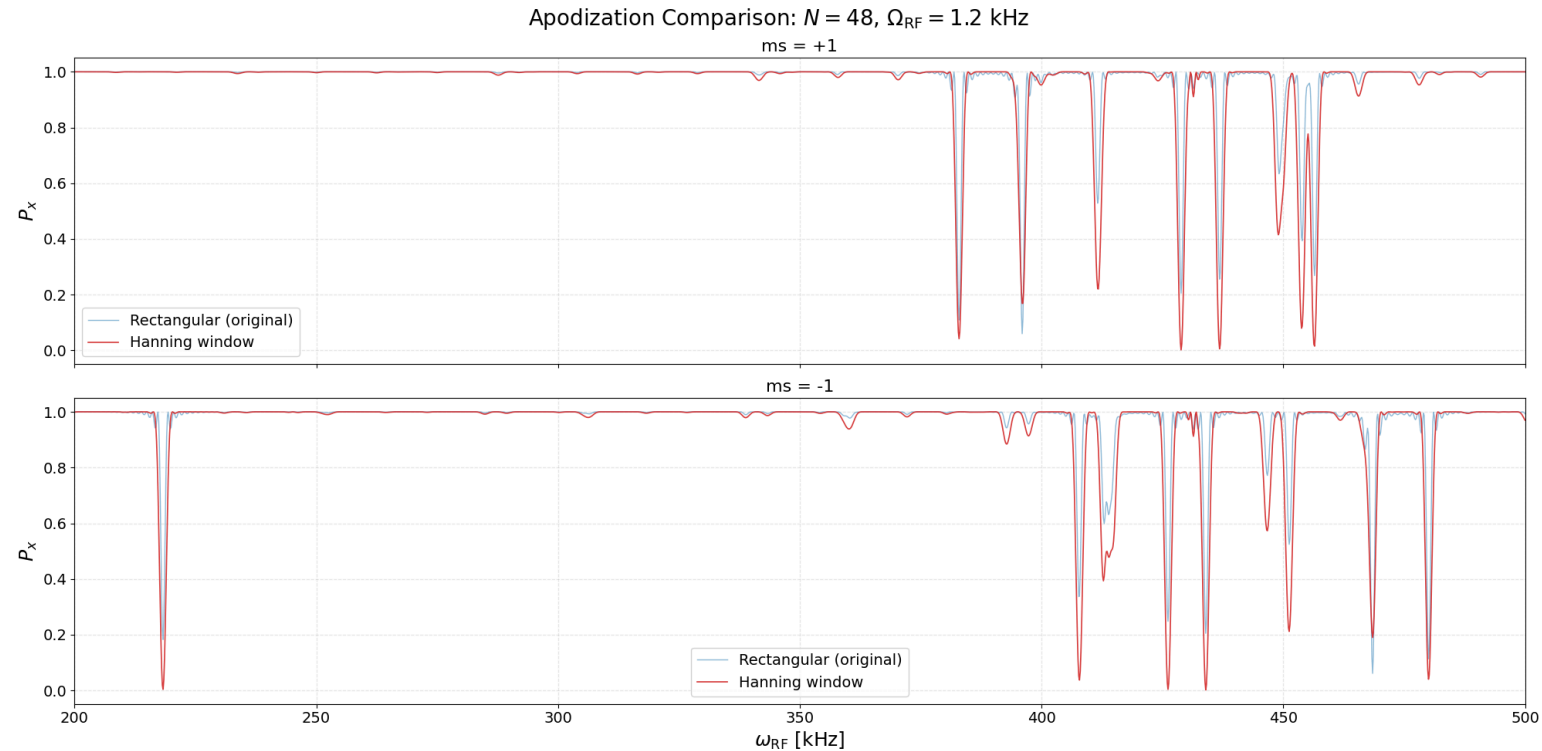
Suppression: Apodized Pulse Idea

Replace constant RF amplitude with a per-cell envelope $\Omega_k = \Omega f(k)$, a discrete window function (Hanning, Hamming, Blackman, ...):



Intuition: the side-lobes are essentially the discrete Fourier transform of a rectangular window. Shaping the window suppresses its sidelobes, exactly the same trick used in classical signal processing.

Suppression: Numerical Result



Apodized envelopes flatten the off-resonant region while preserving the on-resonant peak.

Suppression: Window Comparison

The normalized spectral response factorizes into a window-independent sinc and a window-dependent kernel:

$$\left| \frac{F(\delta_1)}{F(0)} \right|^2 = \text{sinc}^2(u) \cdot |G(u)|^2, \quad u = \frac{\delta_1}{2\Omega \bar{f}}.$$

Window	$\bar{f}$	FWHM (u)	FWHM (δ_1)
Rectangular	1.00	0.89	1.77Ω
Hanning	0.50	1.44	1.44Ω
Hamming	0.54	1.30	1.41Ω
Blackman	0.42	1.68	1.41Ω

Hann/Hamming/Blackman trade broader main lobes for dramatically lower side-lobes.

Outlook: Beyond Constant Ω_{RF}

The full-evolution formalism above **assumes** Ω_{RF} is time-independent (compatible with the RWA). With a Gaussian envelope

$$\Omega_{\text{RF}}(t) = \Omega_0 e^{-(t-t_k)^2/2\sigma^2},$$

the per-segment propagator $e^{-iH'_{0,1}\tau}$ is no longer exact. Direct Schrödinger integration takes tens of minutes per frequency point — impractical for spectroscopy sweeps over thousands of ω_{RF} .

Approach: Magnus expansion, valid when Ω_{RF} varies slowly relative to ω_{RF} .

Outlook: Magnus Expansion

For $\partial_t U = -iH(t)U$, write $U = e^{\Omega(t)}$ with $\Omega(t) = \sum_n \Omega_n(t)$ where

$$\Omega_1 = \int_0^t A_1 dt_1, \quad \Omega_2 = \frac{1}{2} \iint [A_1, A_2] dt_1 dt_2, \dots$$

Computed terms (with $f(t)$ the Gaussian envelope):

$$\Omega_1(T) = -i \left(\delta_{(0,1)} T I_z + c_1 (\cos \phi I_x + \sin \phi I_y) \right)$$

$$\Omega_3(T) = \frac{\delta_{(0,1)}^2}{24} K_1 (\cos \phi I_x + \sin \phi I_y) + \frac{\delta_{(0,1)}}{24} K_2 I_z$$

$$\Omega_2 = \Omega_4 = 0$$

with $c_1 = \int_0^T f$ and $K_{1,2}$ triple integrals of f . Convergence is guaranteed when $\int_0^T \|A(s)\|_2 ds < \pi$ — likely satisfied here; **simulation pending**.

Thank You!

Questions & Discussion